

Predicting the failure of space shuttles (Challenger)

Model: Logistic regression

The Analytics Edge: The risk of launch of space shuttles can be estimated using pre-launch tests and data collected from these tests. Using a logistic regression model, we will estimate the risk of failure of one such space shuttle launch which turned out to be one of the major space disasters. The estimate is made using observable data from pre-launch tests. It helps make predictions for new values of the input, the output of which is unobservable in the original dataset. The model shows that it is important to perform careful risk assessment of systems and subsystems in space programs. This study has effected the way space organizations conduct risk evaluation.

Overview

On January 28, 1986, the NASA space shuttle orbiter Challenger broke apart 73 seconds into its flight, leading to the death of 7 crew members. The disaster resulted in a 32 month hiatus in shuttle programs and the formation of the Rogers Commission to investigate the accident. This included Neil Armstrong and Richard Feynman, a famous physicist along with several other members of prominence. The Commission found that the accident was most probably caused by a failure in the O-ring sealing the aft field joint in one of the booster rockets, causing pressurized hot gases and flames to blow by the O-ring, making contact with the external tank. The Commission also concluded that O-rings did not seal well with low temperature. The weather forecast on January 28, 1986 was 31 degrees Fahrenheit (or -0.55 degrees Celsius). Engineers were concerned that the rubber O-rings were vulnerable to failure at low temperatures. However the space shuttle still took off.

Excerpt from Rogers Report:

The report of the fateful series of NASA and Thiokol meetings, telephone conferences, notes and facsimile transmissions on January 27, the night before the launch of 51-L, shows that only limited consideration was given to the past history of O-ring damage in terms of temperature. The managers compared as a function of temperature, the flights for which thermal distress of O-rings had been observed - not the frequency of occurrences based on all flights. In such a comparison, there is nothing irregular in the distribution of O-ring distress over the spectrum of joint temperatures at launch between 53 and 75 degrees Fahrenheit.

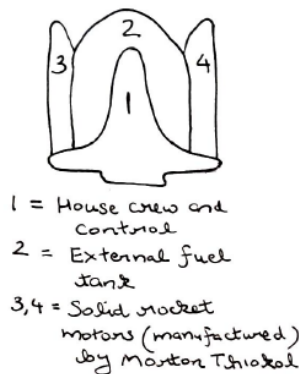


Figure 0.1: Schematic of Challenger shuttle

Key Question: What is the chances of a catastrophic O-ring failure if the Space Shuttle is launched at 31 degrees Fahrenheit?

There were 24 launches prior to the fateful Challenger disaster. For one of the flights, motors were lost at the sea. So motor failure data was available for each of the six O-rings for each of the 23 flights. The data on the temperature and pressure information for these launches is also available.

Summary

Data: The data is a small dataset that provides information on test launches. It is available in the paper - Risk Analysis of the Space Shuttle: Pre-Challenger Prediction of Failure, Journal of the American Statistical Association.

Model: A logistic regression model helps predict the probability of failure of an event using variables that are identified as potential predictors. The output variable here is binary and the model has to account for this explicitly.

Decision: The model provides an estimate of the probability of failure. Based on the risk tolerance, one would decide to launch or not launch the space shuttle using these estimates. It is of course unlikely that in any practical complex system, this number will be exactly zero.

Value: The value of the model is that it provides estimates for new scenarios that one might not see in the original dataset (very low temperatures as in this example). Lives might be saved in such applications.